

2	(a)	The rotating drum exerts a normal reaction force on the clothes and water which acts towards the centre of the rotation. No such force is possible at the holes and so the water over each hole, according to Newton's 1st law, flies off at a tangent to the drum's motion. While the water can pass through the holes in the drum, the clothes are too large to do this, so water is removed from the clothes during the spin-dry cycle. OR The clothes cannot provide sufficient centripetal force for the water, through their mutual attraction, at the holes.	B1 B1 B1
	(b)	$\omega = 1000 \text{ rev per min} = 1000 \times 2\pi / 60 = 105 \text{ rad s}^{-1}$ $F = m r \omega^2 = 0.500 \times 0.125 \times 104.72^2$ $= 685 \text{ N}$	A1 C1 A1
	(c)	When the inner drum is rotating, a purpose of the springs is to prevent damage to the machine / excessive noise by damping the shock of the vibrations OR When the inner drum is not rotating, a purpose of the springs is to	B1 B1

		maintain the drums and clothes as a system in equilibrium position via tension in the top springs and compression in the bottom springs. OR Any other suggestions with explanation based on good physics.	
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